

Emotional Stress State Classification and Analysis using Spiking Neural Networks

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INTRODUCTION

Human affect recognition is a pivotal research area with direct impact on **health & wellbeing**, **human-machine interaction** and **marketing**. Among physiological signals, **electroencephalogram (EEG)** is extensively used to classify brain activity associated with emotions (Bastos-Filho et al., 2012).

Conventional pipelines rely on hand-crafted feature extraction and often neglect **spatial representations** of EEG topography. In this study we introduce a **simple machine-learning model based on Spiking Neural Networks (SNN) with Hebbian-like learning** that operates **global, manual feature extraction** and preserves spike-timing information via Address Event Representation (Farabet et al., 2012).

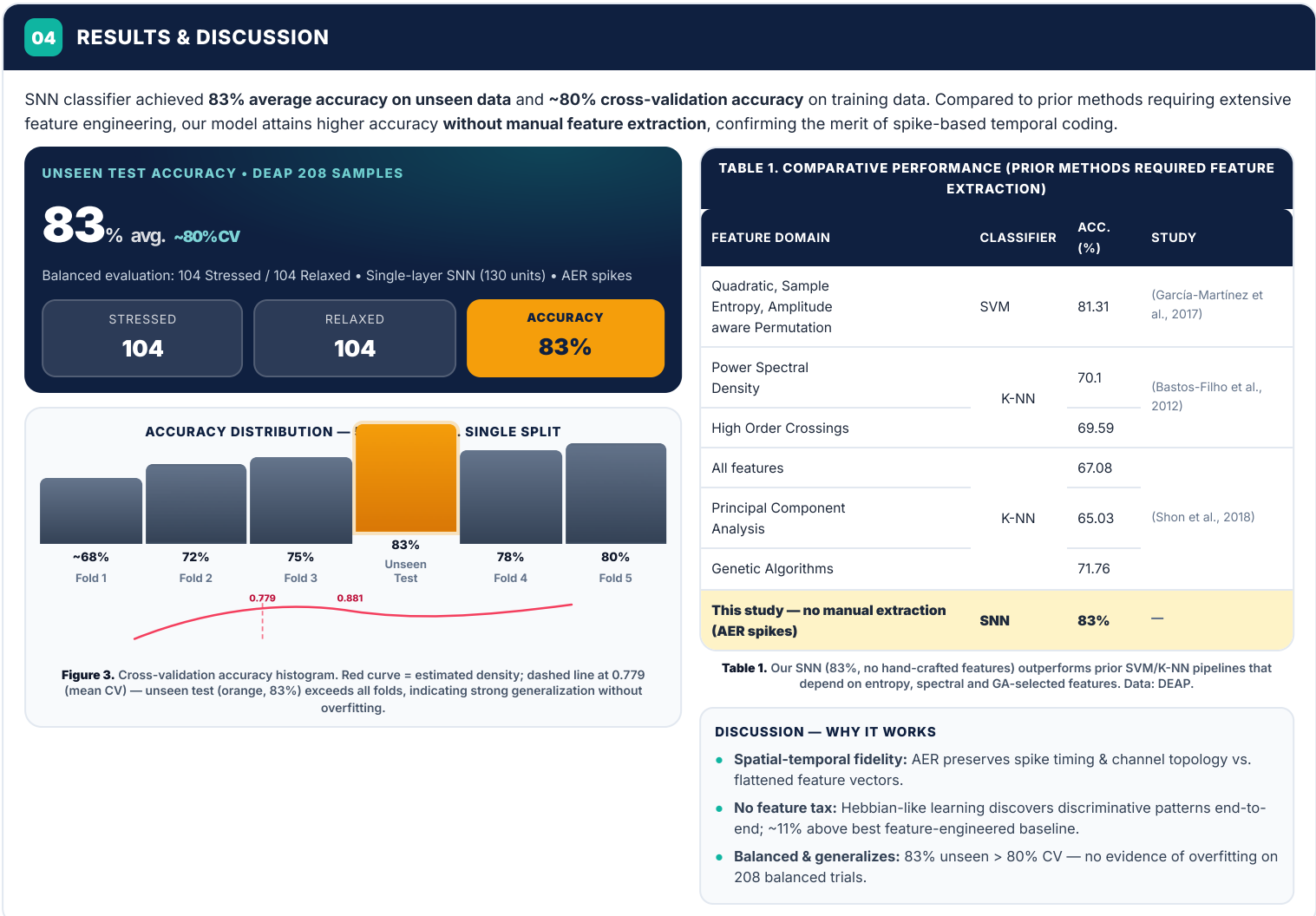
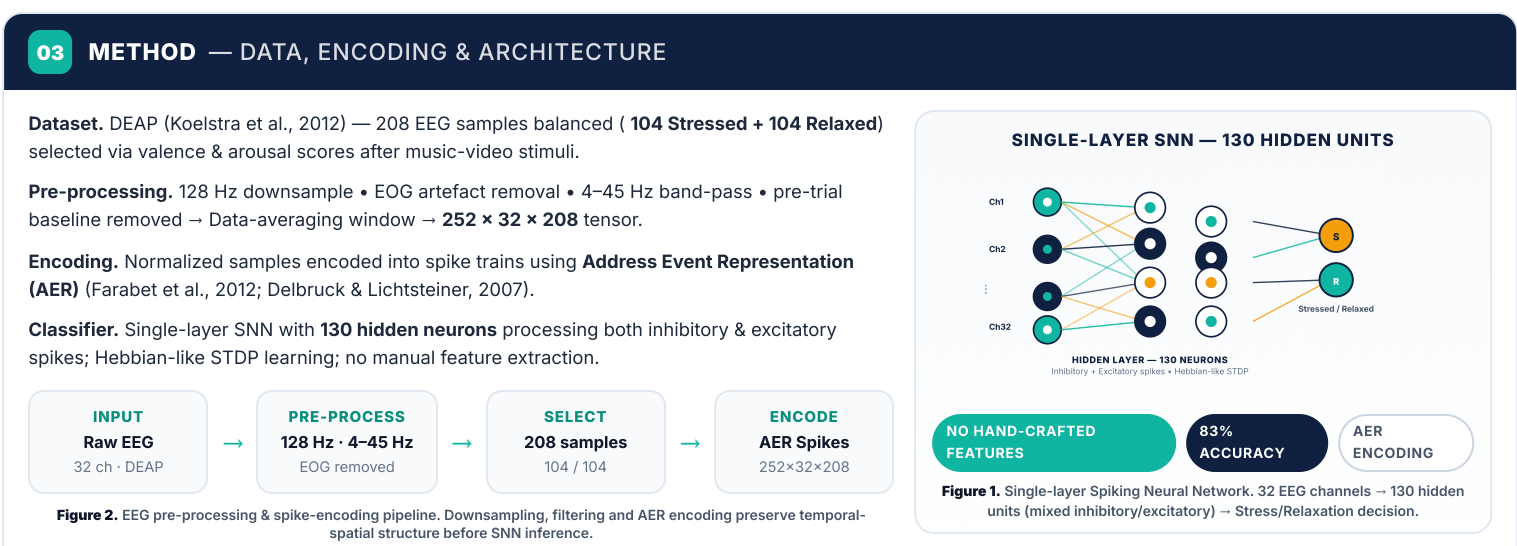
Key contribution: SNN classifier discriminates **Emotional Stress vs. Relaxation** brain states at **83% average accuracy** on unseen data — **state-of-the-art** without manual features.

- OBJECTIVES
- 1 Experiment & evaluate

Benchmark other machine-learning algorithms for EEG-based stress classification against the SNN baseline.
- 2 Optimize SNN performance

Tune topology, learning rates and spike encoding to improve accuracy and reduce training time.
- 3 Integrate into a web platform

Deploy the trained SNN as an interactive web service for real-time stress monitoring.



CONCLUSION

This research delivers a **state-of-the-art 83% accuracy** for Stress vs. Relaxation classification **without manual feature extraction** — simplifying the pipeline and retaining spatial structure.

Future work: implement anti-STDP learning to **reduce SNN training time** and further optimize performance before web-platform deployment.

Next Steps

Anti-STDP — Faster training
+ SNN web platform integration (Q4)
Open for Platform Integration

REFERENCES

Bastos-Filho, T. F., Farias, A., Jansen, A. C., Aguiar, S., & Kumar, D. (2012). Evaluation of feature extraction techniques in emotional state recognition. *Int. Conf. on Intelligent Human Computer Interaction*.
Devlin-Morales, J., & Lachaux, J. P. (2012). Fast online binary vector-based on-line-based hybrid response-projection system. *ICSP*.
Farabet, C., et al. (2012). Comparison between frame-convolution and event-driven visual processing. *Proc. ICAP*.
García-Martín, A., Martínez-Sánchez, A., Sengco, R., Pardo, J. M., & Navarro, J. (2017). Continuous and EEG-based stress classification. *Physiol. Meas.*
Koehnle, S., et al. (2012). DEAP: A database for emotion analysis using physiological signals. *IEEE Trans. Affective Computing*.
Shen, D., et al. (2018). EEG-based emotion recognition using genetically optimized K-NN. *IEEE Access*.